

Sihun Kim

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Senior Software Engineer with 5 years of experience designing and delivering full-stack web platforms. Works across frontend architecture (Vue/Nuxt, TypeScript) and backend systems (Node.js, GraphQL, NestJS), with a focus on scalable data layers and headless CMS integration. Actively exploring vector search and RAG pipelines.

Technical Skills

Frontend: Vue, NuxtJS (SSR/SSG), CSS3, Responsive Design, Accessibility (WCAG), Design Systems

Backend & APIs: Node.js, GraphQL, REST APIs, NestJS

Backend Frameworks (POC): FastAPI, Django Rest Framework

Databases & Search: PostgreSQL, Elasticsearch, Typesense, Qdrant

Cloud & Infrastructure: AWS Lambda, AWS EC2, Serverless Architecture, AWS CDK

Tools: Git, Docker, RabbitMQ, Drizzle, Prisma, pnpm, Vite, esbuild

AI / Search: Vector embeddings, semantic search, RAG pipelines

Professional Experiences

Senior Software Engineer

EdPlus at Arizona State University

Aug 2025 - Present · Scottsdale, AZ

- Redesigned tuition calculator data pipeline, moving dynamic tuition retrieval to request time to reduce build-time API load and improve data accuracy.
- Designed a data resolution layer normalizing inconsistent third-party APIs into structured GraphQL responses, simplifying frontend data consumption.
- Maintain and evolve a Nuxt 3 SSG platform integrated with headless Drupal CMS supporting large-scale educational content delivery.
- Implemented backend data processing and caching strategies improving system reliability.
- Mentored junior engineers during onboarding, establishing knowledge transfer practices that improved ramp-up efficiency.

Software Engineer

EdPlus at Arizona State University

Jan 2022 - Aug 2025 · Scottsdale, AZ

- Led proof-of-concept initiatives evaluating Nuxt 3, Astro, and JAMstack architectures, informing platform modernization decisions.
- Rebuilt backend data layer using GraphQL (Apollo Server on Nuxt Nitro) with filtering and sorting logic to support scalable API design.
- Reorganized program taxonomy and CMS data pipelines to support category-based program types, enabling flexible content querying and display.
- Developed Drupal content extraction pipelines and JSON transformation workflows, enabling reliable CMS-to-frontend data integration.
- Designed and updated Drupal content and paragraph types, improving accessibility and content authoring workflows.

Web Administrator / IT Technician

The Korea Times

Sep 2018 - Apr 2021 · Glenview, IL

- Optimized company website based on user feedback, resulting in 300% improved page loading speed and over 100% page views.
- Created a remote working protocol and data-sharing network that smoothened the transition to virtual work due to COVID-19.
- Collaborated with vendors to acquire software for an interactive e-newspaper, which attracted subscribers and increased ad revenue.

Selected Engineering Projects

Manguito Component Library - SSR-Compatible Vue Design System

github.com/bobbykim89/manguito-component-library

Reusable design system used for rapid prototyping foundation for multiple POC projects.

Tech: Vue3, TypeScript, TailwindCSS, Storybook, Lerna

- Built modular component architecture supporting SSR environments such as Nuxt.
- Designed scalable theming system using Tailwind-based design tokens.
- Published reusable packages to npm and documented components with Storybook.

Vector Search Engine Comparison - Semantic Search Infrastructure POC

github.com/bobbykim89/poc-search-engines

Proof-of-concept project evaluating vector search engines for future product architecture.

Tech: Python, Docker, OpenAI Embeddings, SBERT Embedding models, Qdrant, Elasticsearch, Typesense, Streamlit

- Built Docker environment benchmarking Qdrant, Elasticsearch, and Typesense for semantic similarity search.
- Implemented embedding pipeline generating vectors using multiple embedding models.
- Developed Streamlit UI enabling interactive comparison of search results and similarity scores.

Mango Planner — Offline-first PWA Task Manager

github.com/bobbykim89/mango-planner-v2

Productivity app designed around real-world user feedback and offline reliability.

Tech: Nuxt 3, MongoDB, Workbox, TailwindCSS

- Implemented secure backend API with JWT authentication.
- Built offline-first PWA architecture using Workbox with local draft persistence to prevent data loss during network failures.
- Designed markdown-enabled task editor with math rendering support.

Middle Earth Chat - RAG Knowledge Assistant

github.com/bobbykim89/LLM-Project-LOTR

Retrieval-Augmented Generation chatbot answering lore-based questions using canonical Tolkien texts.

Tech: Python, Qdrant, Jina embeddings, OpenAI API, Django Rest Framework

- Built vector retrieval pipeline using Qdrant and Jina embeddings for semantic search.
- Implemented RAG architecture combining semantic retrieval with LLM response generation.
- Designed RAG evaluation pipeline benchmarking retrieval ranking (MRR on top-k results) and response quality using LLM-as-judge scoring.

Education & Certificates

Purdue University

Bachelor of Science, Industrial Engineering - May 2018

Certifications

AWS Certified Cloud Practitioner - Dec 2023

LLM Zoomcamp, DataTalks Club - Nov 2025

Miscellaneous Information

U.S. Permanent Resident (Green Card holder), U.S. naturalization in progress

Republic of Korea Army - Honorable discharge, Sergeant (mandatory military service)

Languages: English (fluent), Korean (native)

Technical interests: AI applications, vector search, developer tooling, frontend platform architecture, serverless architecture